

Challenging Numerical Questions on Random Forest Regression with Solutions

Ten mathematical and numerical problems covering ensemble prediction, split selection, impurity reduction, bootstrap/OOB concepts, variance, feature randomness, leaf prediction, OOB error, permutation importance, and model comparison.

Question 1: Ensemble Prediction and Squared Error

A Random Forest Regressor contains five regression trees. For a test sample, the individual tree predictions are 42, 50, 47, 55, and 46. The actual target value is 48.

- (a) Calculate the final Random Forest prediction.
- (b) Calculate the squared error of the final prediction.
- (c) Calculate the mean and variance of the five tree predictions.

Solution

Final prediction = $(42 + 50 + 47 + 55 + 46) / 5 = 48$.

Squared error = $(48 - 48)^2 = 0$.

Mean = 48.

Variance = $[(42-48)^2 + (50-48)^2 + (47-48)^2 + (55-48)^2 + (46-48)^2] / 5$
 $= (36 + 4 + 1 + 49 + 4) / 5 = 18.8$.

Answer: Prediction = 48, squared error = 0, variance = 18.8.

Question 2: Comparing Candidate Splits Using Weighted MSE

For an ordered feature X, the data are:

X: 1, 2, 3, 4, 5, 6

Y: 4, 6, 8, 18, 20, 22

Compare two candidate splits:

Split A: $X \leq 2.5$

Split B: $X \leq 3.5$

Calculate the weighted MSE for both splits and identify the better split.

Solution

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Solution

Split A:

Left Y = {4, 6}; mean = 5; MSE = 1.

Right Y = {8, 18, 20, 22}; mean = 17.

Right MSE = $[81 + 1 + 9 + 25] / 4 = 29$.

Weighted MSE = $(2/6)(1) + (4/6)(29) = 19.6667$.

Split B:

Left Y = {4, 6, 8}; mean = 6; MSE = $8/3 = 2.6667$.

Right Y = {18, 20, 22}; mean = 20; MSE = $8/3 = 2.6667$.

Weighted MSE = $(3/6)(2.6667) + (3/6)(2.6667) = 2.6667$.

Answer: Split B is better because it has the lower weighted MSE.

Question 3: Impurity Reduction

A parent node contains target values {5, 7, 9, 15, 17, 19}. A split produces:

Left node = {5, 7, 9}

Right node = {15, 17, 19}

Calculate the parent MSE, weighted MSE after splitting, and impurity reduction.

1. Understanding the Parent Node

Before splitting, all target values are in one group: {5, 7, 9, 15, 17, 19}. This is called the parent node. MSE measures how spread out the target values are.

2. Calculate the Parent Mean

Mean = $(5 + 7 + 9 + 15 + 17 + 19) / 6$
 $= 72 / 6 = 12$

Therefore, the parent mean is 12. If the node is not split, the regression tree predicts 12 for observations in this node.

3. Calculate the Parent MSE

$$\text{MSE} = \sum (Y_i - \text{Mean})^2 / n$$

$$(5 - 12)^2 = 49$$

$$(7 - 12)^2 = 25$$

$$(9 - 12)^2 = 9$$

$$(15 - 12)^2 = 9$$

$$(17 - 12)^2 = 25$$

$$(19 - 12)^2 = 49$$

$$\text{Sum} = 166$$

$$\text{Parent MSE} = 166 / 6 = 27.6667$$

The high MSE means that the target values are widely spread. Impurity means how different or spread out the target values are within a node.

4. Split the Parent Node

The parent node is divided into:

$$\text{Left node} = \{5, 7, 9\}$$

$$\text{Right node} = \{15, 17, 19\}$$

The objective is to create groups containing more similar target values.

5. Calculate the Left Node MSE

$$\text{Left mean} = (5 + 7 + 9) / 3 = 7$$

$$(5 - 7)^2 = 4$$

$$(7 - 7)^2 = 0$$

$$(9 - 7)^2 = 4$$

$$\text{Left MSE} = (4 + 0 + 4) / 3$$

$$= 8 / 3$$

$$= 2.6667$$

The MSE is small because 5, 7, and 9 are close to each other.

6. Calculate the Right Node MSE

$$\text{Right mean} = (15 + 17 + 19) / 3 = 17$$

$$(15 - 17)^2 = 4$$

$$(17 - 17)^2 = 0$$

$$(19 - 17)^2 = 4$$

$$\text{Right MSE} = (4 + 0 + 4) / 3$$

$$= 8 / 3$$

$$= 2.6667$$

The MSE is small because 15, 17, and 19 are close to each other.

7. Calculate the Weighted MSE

$$\text{Weighted MSE} = (nL / n) \times \text{MSE_Left} + (nR / n) \times \text{MSE_Right}$$

Here:

$$nL = 3, nR = 3, n = 6$$

$$\text{Weighted MSE} = (3/6)(2.6667) + (3/6)(2.6667)$$

$$= 1.33335 + 1.33335$$

$$= 2.6667$$

The weight 3/6 is used because 3 out of 6 observations belong to each child node.

8. Calculate Impurity Reduction

$$\text{Parent MSE} = 27.6667$$

$$\text{Weighted MSE} = 2.6667$$

$$\text{Impurity Reduction} = \text{Parent MSE} - \text{Weighted MSE}$$

$$= 27.6667 - 2.6667$$

$$= 25$$

Impurity reduction measures how much the split improves the node by reducing target-value variation.

Simple Interpretation and Final Answer

Before splitting, the values are widely spread, and the MSE is 27.6667.

After splitting, {5, 7, 9} and {15, 17, 19} contain values that are close to each other. The weighted MSE becomes 2.6667.

Final Answer:

Parent MSE = 27.6667
Weighted MSE = 2.6667
Impurity Reduction = 25

In simple words: The split is good because it reduces the target-value variation from 27.6667 to 2.6667. In Random Forest Regression, a split with a larger impurity reduction is generally preferred.

Question 4: Bootstrap Sampling and OOB Prediction

A dataset contains A, B, C, D, and E. Three trees use these bootstrap samples:

Tree 1: {A, B, B, D, E}
Tree 2: {A, A, C, D, D}
Tree 3: {B, C, C, E, E}

- (a) Identify the OOB observations for each tree.
- (b) Which tree can produce an OOB prediction for C?
- (c) If Tree 1 predicts 25 for C and the actual value is 30, calculate the squared OOB error.

Solution

Tree 1 excludes C. OOB set = {C}.
Tree 2 excludes B and E. OOB set = {B, E}.
Tree 3 excludes A and D. OOB set = {A, D}.

Only Tree 1 can provide an OOB prediction for C.

Squared OOB error = $(30 - 25)^2 = 25$.

Question 5: Variance of a Random Forest

Each regression tree has prediction variance $\sigma^2 = 100$. Pairwise correlation between tree predictions is $\rho = 0.2$. A forest contains $B = 10$ trees.

Using:
$$\text{Variance(RF)} = \rho\sigma^2 + [(1-\rho)\sigma^2 / B]$$

Calculate the forest variance and the limiting variance as B approaches infinity.

Solution

$$\begin{aligned}\text{Variance(RF)} &= (0.2)(100) + [(0.8)(100)/10] \\ &= 20 + 8 = 28.\end{aligned}$$

As B approaches infinity, the second term approaches zero.

Limiting variance = $\rho\sigma^2 = (0.2)(100) = 20$.

Answer: Variance for 10 trees = 28; limiting variance = 20.

Question 6: Random Feature Selection

A dataset contains 16 input features. At each split, a Random Forest randomly selects 4 candidate features.

- (a) What percentage of features is examined at each split?
- (b) What is the probability that a particular feature is selected?
- (c) Over 20 independent splits, find the expected number of times a particular feature is considered.

Solution

Percentage = $(4/16) \times 100 = 25\%$.

Probability for a particular feature = $4/16 = 0.25$.

Expected number = $n \times p = 20 \times 0.25 = 5$.

Answer: 25%, probability = 0.25, expected appearances = 5.

Question 7: Terminal Leaf and Forest Prediction

A test observation reaches the following terminal leaves:

Tree 1: {10, 12, 14}

Tree 2: {8, 12}

Tree 3: {15, 17, 19, 21}

Tree 4: {11, 13, 15}

Each tree predicts the mean of its leaf. Find each tree prediction, the Random Forest prediction, and the absolute and squared errors if the actual target is 15.

Solution

Tree 1 prediction = 12.

Tree 2 prediction = 10.

Tree 3 prediction = 18.

Tree 4 prediction = 13.

Forest prediction = $(12 + 10 + 18 + 13)/4 = 13.25$.

Absolute error = $|15 - 13.25| = 1.75$.

Squared error = $(15 - 13.25)^2 = 3.0625$.

Answer: RF prediction = 13.25, absolute error = 1.75, squared error = 3.0625.

Question 8: OOB-MSE and OOB-RMSE

The actual values and OOB predictions are:

Observation 1: Actual 20, OOB prediction 18

Observation 2: Actual 25, OOB prediction 27

Observation 3: Actual 30, OOB prediction 29

Observation 4: Actual 40, OOB prediction 36

Calculate OOB-MSE and OOB-RMSE.

Solution

Squared errors are:

$$(20-18)^2 = 4$$

$$(25-27)^2 = 4$$

$$(30-29)^2 = 1$$

$$(40-36)^2 = 16$$

$$\text{OOB-MSE} = (4 + 4 + 1 + 16)/4 = 6.25.$$

$$\text{OOB-RMSE} = \sqrt{6.25} = 2.5.$$

Answer: OOB-MSE = 6.25 and OOB-RMSE = 2.5.

Question 9: Permutation Feature Importance

A Random Forest has baseline test MSE = 12. After permuting individual features:

Feature A: MSE = 30

Feature B: MSE = 15

Feature C: MSE = 24

Feature D: MSE = 13

Define importance = Permuted MSE - Baseline MSE. Calculate and rank all features.

Solution

$$\text{Feature A} = 30 - 12 = 18.$$

$$\text{Feature B} = 15 - 12 = 3.$$

$$\text{Feature C} = 24 - 12 = 12.$$

$$\text{Feature D} = 13 - 12 = 1.$$

Ranking: A > C > B > D.

Answer: A is most important, followed by C, B, and D.

Question 10: Comparing Two Random Forest Models

Actual values are {50, 60, 70}.

Model A predictions: {48, 65, 68}

Model B predictions: {52, 58, 75}

For each model, calculate MAE, MSE, and RMSE. Identify the better model based on MSE.

Solution

Given Data

MAE is Mean Absolute Error, MSE is Mean Squared Error, and RMSE is Root Mean Squared Error. Lower error values generally indicate better prediction performance.

Model A: Errors

Error = Actual – Predicted

$$50 - 48 = 2$$

$$60 - 65 = -5$$

$$70 - 68 = 2$$

Errors = {2, -5, 2}

The negative error means the model predicted higher than the actual value.

MAE for Model A

Absolute errors = {2, 5, 2}

$$\text{MAE} = (2 + 5 + 2) / 3 = 9 / 3 = 3$$

On average, Model A is 3 units away from the actual values.

MSE for Model A

Squared errors:

$$2^2 = 4$$

$$(-5)^2 = 25$$

$$2^2 = 4$$

$$\text{MSE} = (4 + 25 + 4) / 3 = 33 / 3 = 11$$

MSE gives a larger penalty to larger errors.

RMSE for Model A

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{11} \approx 3.3166$$

Therefore, Model A RMSE is approximately 3.32 units.

Model B: Errors

$$50 - 52 = -2$$

$$60 - 58 = 2$$

$$70 - 75 = -5$$

$$\text{Errors} = \{-2, 2, -5\}.$$

MAE for Model B

$$\text{Absolute errors} = \{2, 2, 5\}$$

$$\text{MAE} = (2 + 2 + 5) / 3 = 9 / 3 = 3$$

On average, Model B is also 3 units away from the actual values.

MSE for Model B

Squared errors:

$$(-2)^2 = 4$$

$$2^2 = 4$$

$$(-5)^2 = 25$$

$$\text{MSE} = (4 + 4 + 25) / 3 = 33 / 3 = 11.$$

RMSE for Model B

$$\text{RMSE} = \sqrt{11} \approx 3.3166$$

Therefore, Model B RMSE is approximately 3.32 units.

Final Comparison

Model A: MAE = 3, MSE = 11, RMSE $\approx$ 3.3166

Model B: MAE = 3, MSE = 11, RMSE $\approx$ 3.3166

Model A has absolute errors {2, 5, 2}, while Model B has {2, 2, 5}. The order of errors differs, but the total absolute and squared errors are the same.

Final Answer: Both Random Forest models perform equally based on MSE because $MSE_A = MSE_B = 11$. Neither model is better based on MAE, MSE, or RMSE.